Catholic Junior College

JC 2 Preliminary Examinations

Higher 1

CHEMISTRY

8873/01

Paper 1 Multiple Choice

16 September 2021

1 hour

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, HT group and NRIC/FIN number on the Answer Sheet in the spaces provided.
Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
The use of an approved scientific calculator is expected, where appropriate.

For each question there are ***four*** possible answers, ***A***, ***B***, ***C*** and ***D***. Choose the **one** you consider to be correct.

1 The relative abundances of the isotopes of a sample of krypton are shown in the table below.

Relative Isotopic Mass	82	83	84	86
Percentage Abundance	7.8	8.6	76.0	7.6

What is the relative atomic mass of krypton in this sample, given to 2 decimal places?

A 83.91 **B** 83.90 **C** 83.89 **D** 83.88

2 Carbon disulfide vapour, $CS_2(g)$, burns in oxygen to give $CO_2(g)$ and $SO_2(g)$.
A sample of 20 cm^3 of $CS_2(g)$ was completely burned in 100 cm^3 of oxygen. The volume of gas after burning was measured. This gaseous mixture was then treated with an excess of KOH(aq) and the volume of gas measured again. All measurements were made at the same temperature and pressure; under such conditions that carbon disulfide was gaseous.
What were the measured volumes?

	Volume of gas after burning/ cm^3	Volume of gas after adding KOH(aq)/ cm^3
A	60	0
B	60	40
C	100	40
D	10	100

3 Which one of the following has the same number of particles as one mole of magnesium atoms?

A the number of hydrogen atoms in 18.0 g of water
B the number of ions in 2 dm^3 of 0.25 mol dm^{-3} of aqueous nitric acid
C the number of delocalised electrons in two moles of copper metal
D the number of molecules in 35.5 g of chlorine gas

4 *Use of the Data Booklet is relevant to this question.*

The ^{68}Ge isotope of the Group 14 element germanium is medically useful because it undergoes a natural radioactive process to give a gallium isotope, ^{68}Ga, which can be used to detect tumors. This transformation of germanium occurs when one of its electrons enters the nucleus, changing a proton into a neutron.

What do the isotopes ^{68}Ga and ^{68}Ge have in common?

A Both isotopes have 37 neutrons in their nuclei.

B Both isotopes have more electrons than protons.

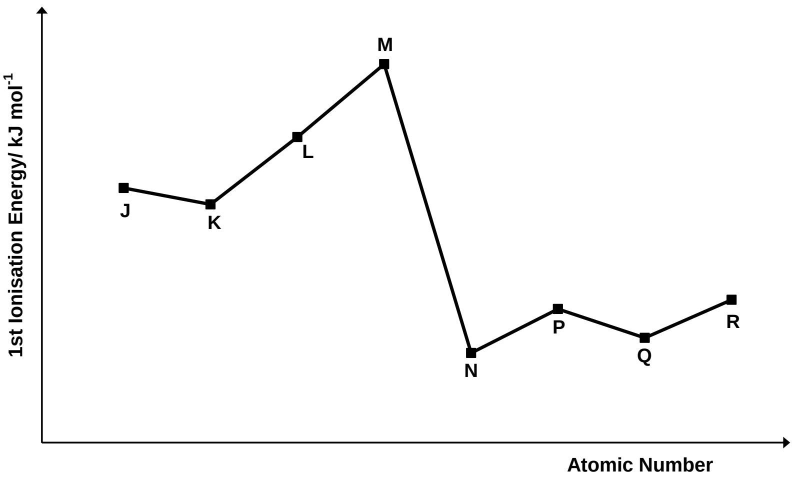
C Both isotopes have an outer electronic configuration $4s^2$ $4p^3$.

D Both isotopes contain the same number of nucleons in their nuclei

5 Which of the following ions will be deflected by the largest angle in an electric field?

A $^{23}Na^{+}$ **B** $^{16}O^{2-}$ **C** $^{27}Al^{3+}$ **D** $^{19}F^{-}$

6 The following graph shows the first ionisation energies of eight consecutive elements **J** to **R**, which have atomic numbers between 3 to 20 in the Periodic Table.



Which one of the following statements about the elements is correct?

A Elements **N** and **K** can form the compound **N_2K**.

B Element **M** is a solid at room temperature.

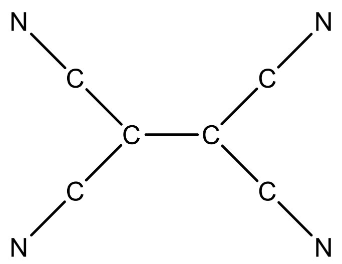
C Aqueous solution of the chloride of element **P** is basic.

D Elements **J** and **R** belong to the same group.

7 *Use of the Data Booklet is relevant to this question.*
Which of the following species has the most number of unpaired electrons?

A Al **B** Si **C** P^{3-} **D** Cr^{3+}

8 Tetracyanoethene is used in the preparation of semiconductors. The structure of tetracyanoethene is given below (double and triple bonds are not shown).



How many sigma, σ, and pi, π, bonds are there in the molecule?

	σ	π
A	9	0
B	9	1
C	9	5
D	9	9

9 Which one of the following sequences shows the bonds arranged in order of increasing polarity?

A	H – O	H – F	H – C*l*
B	H – O	H – C*l*	H – F
C	H – C*l*	H – O	H – F
D	H – C*l*	H – F	H – O

10 Consider the following four molecules.

1 HF
2 I_2
3 H_2O
4 CH_4

What is the order of **increasing** boiling point of the molecules?

A 4 → 1 → 3 → 2
B 4 → 2 → 1 → 3
C 4 → 1 → 2 → 3
D 1 → 4 → 3 → 2

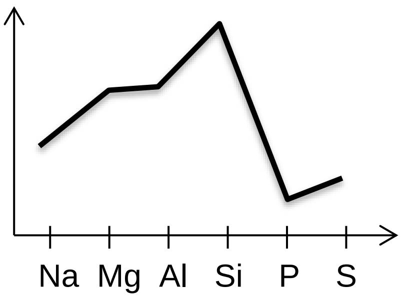
11 Which molecules contain within their structures three atoms arranged in a straight line?

1 SO_2 **2** HCN **3** XeF_4

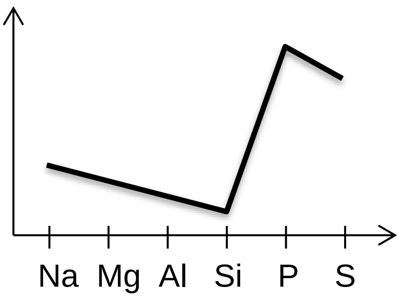
A 2 only **B** 1 and 2 only **C** 1, 2 and 3 **D** 2 and 3 only

12 Which of the following graph shows how electrical conductivity of Period 3 elements varies?

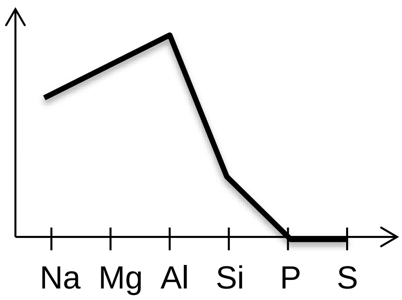
A



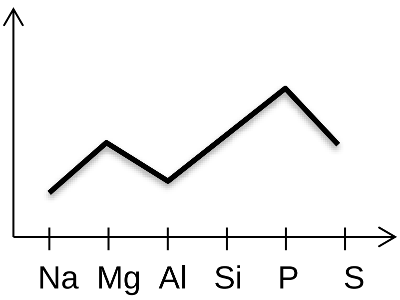
C



B



D



13 The element astatine lies below iodine in Group 17 of the Periodic Table.
Which one of the following properties is correct for astatine?

A It forms diatomic molecules which dissociate less readily than chlorine molecules.
B The hydride of astatine, HAt, is stable to heat.
C It can oxidise iodide to iodine.
D It exists as a crystalline solid at room temperature.

14 Lithium has one valence electron and belongs to Group 1 of the Periodic Table.
Which of the following statements are true about lithium?

1 Lithium has the smallest radius in Group 1.
2 Lithium is the weakest reducing agent in Group 1.
3 Lithium is the most chemically reactive in Group 1.

A 1 only **B** 3 only **C** 1 and 2 only **D** 1, 2 and 3 only

15 Which of the following is an equation that correctly defines a standard enthalpy change of combustion?

A $K(s) + Mn(s) + 2O_2(g) \rightarrow KMnO_4(s)$

B $C(g) + \frac{1}{2}O_2(g) \rightarrow CO(g)$

C $Mg(s) + \frac{1}{2}O_2(g) \rightarrow MgO(s)$

D $2Fe^{3+}(g) + 3O^{2-}(g) \rightarrow Fe_2O_3(s)$

16 The following table lists some $\Delta H_f^\ominus$ values.

Compound	$\Delta H_f^\ominus$ / kJ mol^{-1}
$NO_2(g)$	+33.2
$N_2O_5(g)$	+5.0

The enthalpy change for the dissociation of $N_2O_5(g)$ is given below.

$$N_2O_5(g) \rightarrow NO_2(g) + NO_3(g) \qquad \Delta H^\ominus = +92.5 \text{ kJ mol}^{-1}$$

What is the standard enthalpy change of formation of $NO_3(g)$?

A - 130.7 kJ mol^{-1} **C** + 64.3 kJ mol^{-1}

B - 64.3 kJ mol^{-1} **D** + 130.7 kJ mol^{-1}

17 I_2 vapour is a purple gas while H_2 and HI are colourless gases.
An equilibrium is established between I_2, H_2 and HI in a closed vessel.

$$I_2(g) + H_2(g) \rightleftharpoons 2HI(g) \qquad \Delta H_r = -9.2 \text{ kJ mol}^{-1}$$

Which row describes the effects of changing conditions on the colour of the equilibrium mixture above?

	increasing the pressure	decreasing the temperature
A	no change in the colour intensity	colour becomes lighter
B	colour becomes darker	no change in the colour intensity
C	colour becomes lighter	colour becomes darker
D	no change in the colour intensity	colour becomes darker

18 The following equilibrium exists in a sample of aluminium chloride vapour.

$$Al_2Cl_6\,(g) \rightleftharpoons 2AlCl_3\,(g) \qquad \Delta H = +63 \text{ kJ mol}^{-1}$$

Which of the following will increase when the pressure is increased at constant temperature?

A activation energy
B amount of Al_2Cl_6 formed
C enthalpy change of reaction
D rate constant for the forward reaction

19 Water is considered 'hard' when it is high in dissolved minerals such as calcium and magnesium. Hardness in tap water can be determined by titrating a sample of the water against a reagent which reacts with dissolved metal ions. The indicator for this titration requires the pH to be maintained above 7 at 25°C.
Which substances, in aqueous solution, could be used to maintain the pH above 7?

A ammonia only
B ammonia and sodium hydroxide
C ammonia and ammonium chloride
D sodium hydroxide and sodium ethanoate

20 The pH of human blood is constant at about 7.40.
Which ion or molecule present in the human body is responsible for removing H^+(aq) ions from the blood to keep the pH almost constant?

A NH_3 **B** HCO_3^- **C** CO_3^{2-} **D** PO_4^{3-}

21 The decomposition of NO_2 is a second order reaction.

$$2NO_2(g) \rightarrow 2NO(g) + O_2(g)$$

Which one of the following graphs is a straight line?

A $[NO_2]$ against (rate of reaction)^2

B $[NO_2]^2$ against (rate of reaction)

C $[NO_2]$ against (time)^2

D $[NO_2]^2$ against (time)

22 An enzyme, ***alcohol dehydrogenase***, can be used to oxidise ethanol but the enzyme is never used to oxidise butan-1-ol. Why is this so?

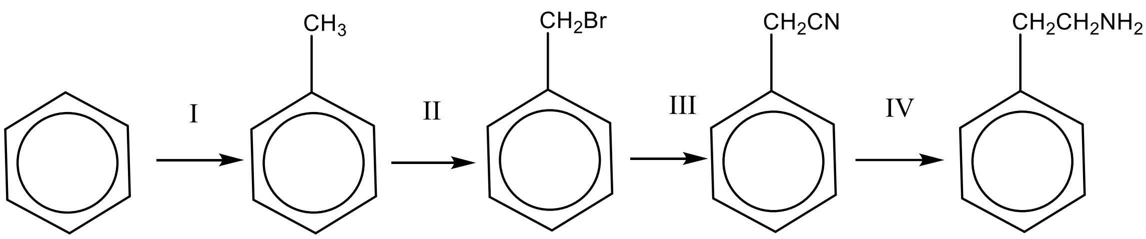
A Ethanol can be oxidised easily whereas butan-1-ol is resistant to oxidation.

B The enzyme is specific to the oxidation of ethanol and is not able to oxidise butan-1-ol.

C There was sufficient enzyme to oxidise ethanol, but more enzyme is required before the oxidation of butan-1-ol occurs.

D Ethanol is easily oxidised at room temperature whereas butan-1-ol requires heat before oxidation occurs.

23 2-phenylethylamine is the 'feel-good' factor in chocolate. It can be synthesised from benzene in the following sequence.



2-phenylethylamine

Which set of information is true of the steps in the sequence?

	I Type of reaction	II Reagents and conditions	III Type of reaction	IV Type of reaction
A	substitution	Br_2, uv light	substitution	reduction
B	substitution	HBr, heat	addition	addition
C	addition	Br_2, uv light	substitution	reduction
D	addition	HBr, heat	addition	addition

24 Which one of the following structural formulae exists in both *cis* and *trans* forms?

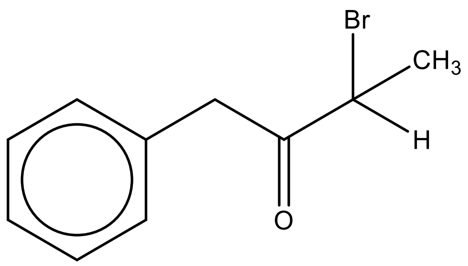
A $CH_2{=}CHCl$

B cyclohexene

C $CHCl{=}C(CH_2CH_3)_2$

D $CH_3CH{=}CHCl$

25 What are the functional groups present in the molecule shown below?



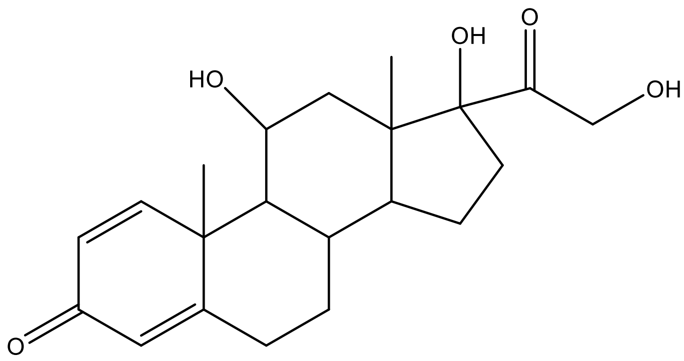
A phenyl, aldehyde and tertiary alkyl bromide

B phenyl, aldehyde and secondary alkyl bromide

C phenyl, ketone and tertiary alkyl bromide

D phenyl, ketone and secondary alkyl bromide

26 *Prednisolone* is a steroid medication used to treat certain types of allergies.



Prednisolone

Which of the following is true of *Prednisolone*?

1 It has a carboxylic acid functional group.

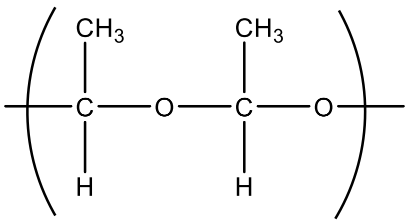
2 It decolourises aqueous bromine.

3 It can undergo esterification upon reaction with CH_3CO_2H.

4 It will not react with $NaBH_4$ in methanol.

A 2 only **B** 2 and 4 only **C** 1 and 3 only **D** 2 and 3 only

27 Ethanal, M_r = 44, can be polymerised in the presence of a suitable reagent. The polymer has the following structure:



Which statements are correct?

1 The relative molecular mass of the polymer molecule is an exact multiple of 44.

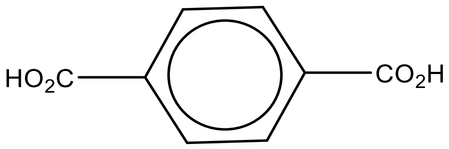
2 The polymer formed is a polyester.

3 The C-C-O bond angle changes from 120º in the monomer to 109º in the polymer.

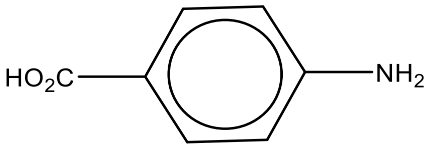
A 1 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

28 Which compound can form a condensation polymer by reacting with itself?

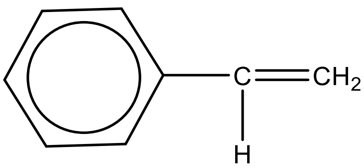
A



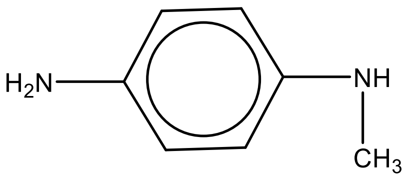
B



C



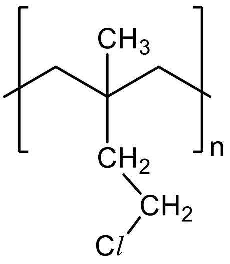
D



29 Superabsorbent polymers (SAPs) are used to create hydrogels that are commonly used in baby diapers. Attempts were even made to use these hydrogels to plug the highly radioactive water leak after the Fukushima nuclear accident. These polymers have the ability to absorb large amounts of water and hold onto the water molecules for an extended period of time.

Which of the following is a likely polymeric structure of such SAPs?

1



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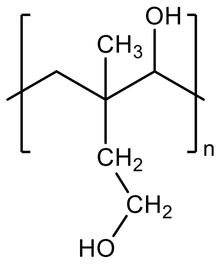

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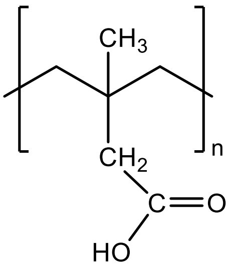

A 1 only　　**B** 1 and 3 only　　**C** 2 and 3 only　　**D** 1, 2 and 3

30 Polymers are used extensively in our daily lives. Various polymers have found their way into products such as:

- soft contact lenses
- wrinkle free shirts
- microwavable food packaging
- bottles for alkaline cleaning solutions

Which row best describes the most suitable polymer to use for each type of product?

	Soft contact lenses	wrinkle free shirts	microwavable food packaging	bottles for alkaline cleaning solutions
A	poly(vinyl alcohol)	polyamide	LDPE	PET
B	poly(vinyl chloride)	polyester	poly(propene)	HDPE
C	poly(vinyl chloride)	polyamide	LDPE	PET
D	poly(vinyl alcohol)	polyester	poly(propene)	HDPE